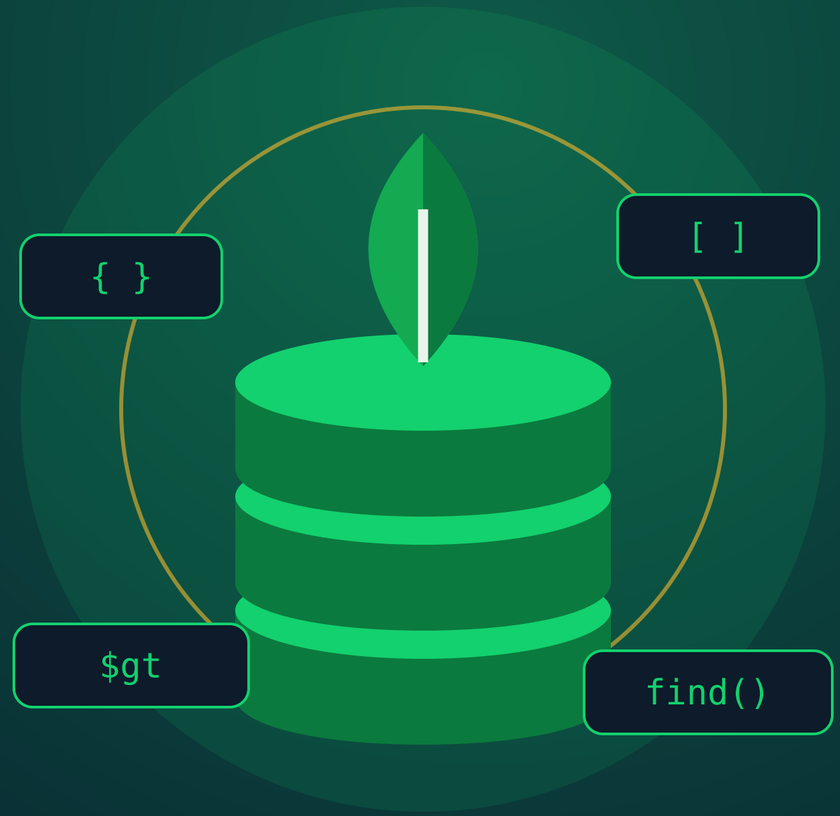
BIG DATA COURSE

SECTION 4

MongoDB Shell Commands Quick Reference

Every command from the section, with what it does. Use it to study.



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SECTION 4

CRUD at a glance

Create

add new data

```
insertOne()
insertMany()
```

Read

search the data

```
find()
findOne()
```

Update

change the data

```
updateOne()
updateMany()
```

Delete

remove the data

```
deleteOne()
deleteMany()
```

Every command has the same shape: `db.collection.operation( filter, change )`. Example: `db.students.updateOne({ name: "Omar Said" }, { $set: { gpa: 3.0 } })`

SECTION 4

Shell basics

1 · START

Command	What we use it for
`mongosh`	Open the MongoDB shell from a terminal
`show dbs`	List the databases that contain data
`use MET`	Switch to database MET (created when you save the first document)
`db`	Show which database you are in
`show collections`	List the collections inside the current database
`help`	Show the shell help
`cls`	Clear the screen
`exit`	Leave the shell

SECTION 4

Create: insert and manage

2 · CREATE

Command	What we use it for
`db.students.insertOne({ ... })`	Add one document
`db.students.insertMany([ { ... }, { ... } ])`	Add many documents in one command
`db.students.countDocuments()`	Count the documents in a collection
`db.students.countDocuments({ major: "IT" })`	Count only the documents that match a filter
`db.createCollection("books")`	Create an empty collection
`db.students.drop()`	Delete the whole collection (no undo)
`db.dropDatabase()`	Delete the current database (no undo)

SECTION 4

Read: find

3 · READ

Command	What we use it for
`db.students.find()`	Show all the documents
`it`	Show the next batch (the shell shows 20 at a time)
`db.students.find({ major: "IT" })`	Find the documents that match a filter
`db.students.find({}, { name: 1, _id: 0 })`	Projection: choose the fields to show (1 shows, 0 hides)
`db.students.findOne({ name: "Sara Ahmed" })`	Return only the first match
`db.students.find({ "address.city": "Cairo" })`	Search inside a nested document (dot notation, quotes required)
`db.students.find({ courses: "SQL" })`	Search inside an array: the array contains SQL
`db.students.find({ courses: ["Networks"] })`	The array is exactly this list

SECTION 4

Read: sort, limit, skip

3 · READ

Command	What we use it for
`.sort({ gpa: -1 })`	Order the results: 1 low to high, -1 high to low
`.limit(3)`	Return only the first 3 results
`.skip(3)`	Skip the first 3 results (used for pagination)
`db.students.find().sort({ name: 1 }).skip(3).limit(3)`	Chain them: page 2 with 3 students per page

Comparison operators

Command	What we use it for
`{ gpa: { $gt: 3 } }`	$gt: greater than
`{ gpa: { $gte: 3.5 } }`	$gte: greater than or equal
`{ age: { $lt: 22 } }`	$lt: less than
`{ age: { $lte: 22 } }`	$lte: less than or equal
`{ major: { $ne: "IT" } }`	$ne: not equal
`{ major: { $in: ["IT", "Data Science"] } }`	$in: the value is one of the list
`{ "address.city": { $nin: ["Cairo"] } }`	$nin: the value is not in the list

Logic and array operators

Command	What we use it for
`{ $or: [ { major: "IT" }, { gpa: { $gte: 3.7 } } ] }`	$or: match any one of the conditions
`{ $and: [ { major: "IT" }, { gpa: { $lt: 3 } } ] }`	$and: match all the conditions
`{ major: "IT", gpa: { $lt: 3 } }`	Several fields in one filter are already AND
`{ courses: { $all: ["Big Data", "AI"] } }`	$all: the array contains all these values
`{ scores: { $elemMatch: { lab: "L2", mark: { $gte: 8 } } } }`	$elemMatch: one array item must match every condition

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Update

5 · UPDATE

Command	What we use it for
`db.students.updateOne(filter, update)`	Change the first matching document
`db.students.updateMany(filter, update)`	Change every matching document
`{ $set: { gpa: 3.0 } }`	$set: change a field or add a new one
`{ $inc: { age: 1 } }`	$inc: add a number to a field
`{ $unset: { graduated: "" } }`	$unset: remove a field

SECTION 4

Update arrays and delete

5 · UPDATE · 6 · DELETE

Command	What we use it for
`{ $push: { courses: "Cloud" } }`	$push: add an item to an array
`{ $push: { courses: { $each: ["Linux", "Docker"] } } }`	$each: add several items at once
`{ $pull: { courses: "Linux" } }`	$pull: remove an item from an array
`{ $addToSet: { courses: "AI" } }`	$addToSet: add only if it is not there yet
`db.students.deleteOne(filter)`	Delete the first matching document
`db.students.deleteMany(filter)`	Delete every matching document
`db.students.deleteMany({})`	Delete ALL documents in the collection (no undo)

SECTION 4

Indexes

7 · SPEED

Command	What we use it for
`db.students.createIndex({ gpa: 1 })`	Create an ascending index (-1 for descending)
`db.students.getIndexes()`	List the indexes of a collection
`db.students.find({ gpa: 3.9 }).explain("executionStats")`	Show how the query ran: IXSCAN uses an index, COLLSCAN reads every document
`db.students.dropIndex("gpa_1")`	Delete an index

Terminal commands (outside mongosh)

Command	What we use it for
`mongosh --version`	Check that the MongoDB shell is installed
`node --version`	Check that Node.js is installed
`npm init -y`	Create the package.json file for a new project
`npm install express mongodb`	Install the web framework and the MongoDB driver
`node index.js`	Start the API server (stop it with Ctrl + C)

BIG DATA COURSE

SECTION 4

Keep practicing

Open mongosh and try every command on the 8 students.

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